Israel Alcantara

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Computer science graduate and MBA student focused on technical project management and IT.

Experience

Amazon Web Services (AWS) – Amazon Dedicated Cloud (ADC), BlackMirror — Arlington, VA
Software Development Engineer Intern — *May 2025 – Aug. 2025*

- Shipped a lossless consecutive-duplicate datapoint compression algorithm into the BlackMirror commercial metrics pipeline (Java, XML), modifying a **shared ADC-wide XML schema** to add a `duplicates` field and delivering a matching decompressor that guarantees exact reconstruction.
- Reduced metric export payloads by **up to 60 GB/day** (avg 38 GB/day) across **3 air-gapped regions**, collapsing up to 100M consecutive duplicate datapoints per region daily while preserving min/max/avg/sum/sample-count fidelity.
- Instrumented the compression path with 13 custom CloudWatch metrics (compression-opportunity flags, bytes saved, duplicate run length, per-metric-type stats) and published a cross-region dashboard used by the wider team to size impact.
- Authored 30+ JUnit tests covering round-trip correctness, edge cases, and error handling; scoped delta-encoding and sum-only compression as follow-on roadmap items for future quarters.

Society of Hispanic Professional Engineers (SHPE), NJIT Chapter — Newark, NJ
Event Coordination Director — *Mar. 2024 – May 2025*

- Organized and executed 12+ networking and professional-development events, coordinating a team of roughly 10 volunteers with maintained checklists and runbooks.
- Managed end-to-end logistics (room bookings, vendor scheduling, AV setup, and corporate-partner coordination) and built reusable planning templates that reduced per-event prep overhead.

New Jersey Institute of Technology — Newark, NJ
Community Assistant, Office of Residential Life — *June 2025 – May 2026*

- Supervise a **400-resident Greek Village complex** across multiple buildings; enforce university housing policy, conduct regular rounds, and author formal incident reports for the Office of Residential Life.
- Primary responder to a late-night guest-policy incident across two buildings; with a graduate CA, a peer CA, and the on-duty Residence Coordinator, de-escalated an uncooperative resident, issued a formal warning, and enforced a building-wide no-entry directive.

Media and Technology Support Services (MTSS), NJIT — Newark, NJ
Delivery Technician & Lab Technician — *Sept. 2022 – May 2026*

- Troubleshoot classroom and lab AV systems end-to-end (projectors, Crestron control systems, AV switchers, wireless microphones and transmitters, classroom network connectivity) across **100+ lectures annually**; systematic diagnosis cut repeat-call rate on the same equipment by an estimated 30%.
- Collaborated with a team of 5 technicians to standardize equipment-setup procedures and pre-lecture readiness checks, improving estimated equipment reliability by 40%; aggregated from daily, weekly, and monthly service-call logs.

Projects

LinkdUp | *React Native, TypeScript, Node.js, Supabase Postgres, Render, Vercel* | linkdup.app — Feb. 2026 – May 2026

- Shipped a production mobile-first web app for group meetup coordination as **backend engineer on a five-person team** for the NJIT CS491 senior capstone, **placing 2nd overall**; deployed live with Google OAuth, Supabase Postgres (27+ schema migrations, row-level security), and Supabase Realtime websockets powering a locked-step swipe-voting engine synchronized across all party members.
- Integrated Google Places API for geographic midpoint venue discovery and Google Calendar API for one-click event export; built a TikTok-style social feed, followers/block system, and collaborative date-voting flow across a full Expo/React Native Web frontend, Express/Node backend, and Render + Vercel deployment pipeline.

SFort95 Compiler | *C++, Git* — Mar. 2024 – Apr. 2024

- Built a full SFort95 compiler in C++ across three pipeline stages: a state-based lexical analyzer producing token reports, a recursive-descent parser with operator-precedence handling, and an interpreter executing the parsed AST with Fortran95-compliant semantics.
- Engineered runtime checks for undefined-variable, type-mismatch, and division-by-zero errors; validated compliance through targeted test cases covering edge grammar productions.

Education

Montclair State University – Feliciano School of Business — Montclair, NJ
Master of Business Administration — *Expected Fall 2028*

Relevant Coursework: Global Economy, Business Statistics, Strategic Information Systems, Business Essentials

New Jersey Institute of Technology – Ying Wu College of Computing — Newark, NJ
Bachelor of Science in Computer Science | GPA: 3.4 — *May 2026*

Relevant Coursework: Operating Systems, Computer Networks, Database System Design, Intensive Programming in Linux, Intro to Cybersecurity, Intro to Machine Learning

Honors: Cum Laude; Dean's List (multiple semesters); Theodore & Lois John, Guastaferro Memorial, and Orsted Opportunity Scholarships

Affiliations: ColorStack Fellow; Hispanic Association for Computing & Computer Sciences (HACCS); Society of Hispanic Professional Engineers (SHPE); Association of Latino Professionals For America (ALPFA) | **Languages**: Fluent in English and Spanish

Technical Skills

Tools, Cloud & Methodologies: Jira, Node.js, Express, Agile/Scrum, REST APIs, AWS (CloudWatch), Linux, Git, XML, JUnit
Databases & Data: Supabase (Postgres, Realtime, Auth), MySQL, MongoDB, Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, Seaborn
Languages: Java, Python, C/C++, Bash/Shell, SQL, JavaScript, TypeScript, HTML, CSS, Kotlin